

DaTra: Data Transformation Semantics as a Symmetric Monoidal Category on Atlas Presheaves

@qualiascript

ChatGPT

Abstract

This work formalizes the semantics of the Data programming-language project in Lean. Data domains are organized into atlases: structured, eventually stable families of pages encoding how data is partitioned and related. Data transformations are modeled as presheaves on the resulting atlas categories, yielding a topos-theoretic semantics. Stable transformations are then assembled over atlas federations, whose horizontal sum extends by Day convolution to a symmetric monoidal category.

1 Dominions

Definition 1.1 (The Category of Dominions). The **Category of Dominions**, denoted $\mathbf{Dom}$, has as its objects sets whose cardinality is at most ω_0 , equivalently sets admitting an injection into ω_0 , and as its morphisms set-theoretic functions.

Definition 1.2 (The Domanial Embedding Functor). The **Domanial Embedding Functor**, denoted $\mathbf{Emb} : \mathbf{Dom} \rightarrow \mathbf{Set}$, sends each dominion to its underlying set.

2 Domanial Insertions

Definition 2.1 (The Category of Domanial Insertions). The **Category of Domanial Insertions**, denoted $\mathbf{DomIns}$, has dominions as objects and monomorphisms in $\mathbf{Dom}$ as morphisms.

Definition 2.2 (The Category of Codomanial Insertions). The **Category of Codomanial Insertions**, denoted $\mathbf{CoDomIns}$, is the opposite category $\mathbf{DomIns}^{\mathrm{op}}$.

3 Chains

Definition 3.1 (Chain). A **Chain** is a totally ordered thin category, indexed skeletally by an ordinal strictly smaller than $\omega_0^{\omega_0}$.

Definition 3.2 (The Spine). The **spine** is the chain with ω_0 objects.

4 Consolidations

Definition 4.1 (The Category of Consolidations). The **Category of Consolidations**, denoted $\mathbf{Con}$, has chains as objects and point-surjective functors between their thin categories as morphisms.

Definition 4.2 (The Category of Coconsolidations). The **Category of Coconsolidations**, denoted $\mathbf{CoCon}$, is the opposite category $\mathbf{Con}^{\mathrm{op}}$.

5 Transportation and Folios

Definition 5.1 (The Transportation Functor). The **Transportation Functor**, denoted $\text{Tra} : \text{Con} \rightarrow \text{Set}$, sends a chain to its object set and a consolidation to its underlying set-theoretic surjection.

Definition 5.2 (Folio). A **folio** is a functor $F : S \rightarrow \text{CoCon}$, where S is the spine, such that $F(0)$ is the singleton chain.

6 Paginations

Definition 6.1 (The Category of Paginations). The **Category of Paginations**, denoted Pag , has as objects pairs (H, E) where $H = \text{Tra} \circ F^{\text{op}}$ for a folio F , and E is the category of elements of H . For $W : \text{Pag}$, write W_H for the first inclusion and W_E for the second inclusion. A morphism $T : X \rightarrow Y$ in Pag is a functor $T : X_E \rightarrow Y_E$. Thus, for every morphism $f : x \rightarrow y$ of X_E , the following square commutes:

$$\begin{array}{ccc} x = (m, i) & \xrightarrow{f} & (n, X_H(f)(i)) \\ T \downarrow & & \downarrow T \\ (m', i') & \xrightarrow{T(f)} & (n', Y_H(T(f))(i')). \end{array}$$

7 Atlases

Definition 7.1 (The Category of Atlases). The **Category of Atlases**, denoted Atl , has as objects pairs (P, G) , where $P : \text{Pag}$ and $G : P_E \rightarrow \text{DomIns}$ is a functor, equivalently a presheaf $G : P_E^{\text{op}} \rightarrow \text{CoDomIns}$. For $P_H : C \rightarrow \text{Set}$, $m \in |C|$, and distinct elements $k, k' \in |P_H(m)|$, the pullback in G of the arrows induced by $(m \rightarrow 0)(k)$ and $(m \rightarrow 0)(k')$ is empty.

For every $n \in \mathbb{N}$, the n **th atlas page** of X is the set $X_H(n)$ on its full spine. An element $k \in X_H(n)$ is a **page cell**, written (n, k) , and the dominion carried by that cell is $X_G(n, k)$. Thus a page is the indexed collection of cells at one spine position.

There is an integer w such that, for every $w' > w$, $P_H(w' \rightarrow w) = \text{Id}$, and, for every $k \in |P_H(w)|$, the corresponding arrow $G((w' \rightarrow w)(k))$ is the identity. The least such integer is the **cardinality of the atlas**, denoted $|A|$.

For $X : \text{Atl}$, write X_P and X_G for the two components and put $X_H = (X_P)_H$ and $X_E = (X_P)_E$. A morphism $T : X \rightarrow Y$ is a pair (T_P, T_A) , where $T_P : X_P \rightarrow Y_P$ is a pagination morphism and $T_A : X_G \Rightarrow Y_G \circ T_E$ is a natural transformation. Thus, for every $f : x \rightarrow y$ in X_E , the following square commutes:

$$\begin{array}{ccc} X_G(x) & \xrightarrow{X_G(f)} & X_G(y) \\ (T_A)_x \downarrow & & \downarrow (T_A)_y \\ Y_G(T_E(x)) & \xrightarrow{Y_G(T_E(f))} & Y_G(T_E(y)). \end{array}$$

Finally, if $x = |X|$, $x' > x$, and $r \in |X_H(|X| - 1)|$, then $T_P(x', r) = T_P(x, r)$ and $(T_A)_{(x', r)} = (T_A)_{(x, r)}$.

Definition 7.2 (Extent of an Atlas). The **extent** of A is $\text{Ex}(A) = A_G(0, 0)$. Since objects of DomIns are dominions, $\text{Ex}(A) : \text{Dom}$ for every $A : \text{Atl}$.

Definition 7.3 (Territory of an Atlas). Let $M = A_H(|A| - 1)$. The **territory** is the indexed family $\text{Ter}(A) : M \rightarrow \text{Dom}$ given by $\text{Ter}(A)(k) = A_G(|A| - 1, k)$.

Definition 7.4 (*n*th Region of an Atlas). The *n*th **region** of A is $\text{Ter}(A)(n)$, with indexing beginning at 0.

8 Transposals and Traversals

Definition 8.1 (The Category of Atlas Transposals). The **Category of Atlas Transposals**, denoted AtlTrap , is the wide subcategory of Atl whose morphisms $F : X \rightarrow Y$ have point-injective object maps F_E .

Definition 8.2 (The Category of Atlas Traversals). The **Category of Atlas Traversals**, denoted AtlTrav , is the wide subcategory of AtlTrap whose morphisms satisfy the following conditions. For $F : X \rightarrow Y$, let $x = (m, i)$ and $y = (m', j)$, put $p = \min(m, m')$, write $F_E(x) = (n, i')$ and $F_E(y) = (n', j')$, and put $p' = \min(n, n')$. If

$$X_H(m \rightarrow p)(i) < X_H(m' \rightarrow p)(j),$$

then

$$Y_H(n \rightarrow p')(i') < Y_H(n' \rightarrow p')(j').$$

Furthermore, if $q \in |X|$ and $t \in X_G(q)$ is in the image of a final region—that is, there exist $k \in |\text{Ter}(X)|$ and $l \in \text{Ter}(X)(k)$ whose image under $X_G(|X| - 1 \rightarrow q)(k)$ is t —then this property is preserved by F .

Definition 8.3 (The Category of Stable Atlas Traversals). The **Category of Stable Atlas Traversals**, denoted StaAtlTrav , is the wide subcategory of AtlTrav whose morphisms $F : X \rightarrow Y$ satisfy $F_E(0, 0) = (0, 0)$.

9 Maps and Cartography

Definition 9.1 (The Category of Atlas Maps). The **Category of Atlas Maps**, denoted AtlMap , is the full subcategory of atlases for which every element of the extent is in the image of a final region.

Definition 9.2 (The Atlas Map Inclusion Functor). The **Atlas Map Inclusion Functor** $\text{AtlMapInc} : \text{AtlMap} \rightarrow \text{Atl}$ is the canonical inclusion.

Definition 9.3 (The Category of Atlas Traversal Maps). The **Category of Atlas Traversal Maps**, denoted AtlTravMap , is the full subcategory of AtlTrav whose objects are Atlas Maps.

Definition 9.4 (The Atlas Traversal Map Inclusion Functor). The canonical inclusion is denoted $\text{AtlTravMapInc} : \text{AtlTravMap} \rightarrow \text{AtlTrav}$.

Lemma 9.5 (Cartography Lemma). *The inclusion AtlTravMapInc has a right adjoint, denoted the **Charting Functor** $\text{Chr} : \text{AtlTrav} \rightarrow \text{AtlTravMap}$.*

Proof sketch. On an object $X : \text{AtlTrav}$, the functor Chr sends X to the universal arrow from AtlTravMapInc to X . Write $X' = \text{AtlTravMapInc}(\text{Chr}(X))$, together with $H : X' \rightarrow X$, with X' terminal among objects having this property. The solution is the DaTra map satisfying $\text{Ter}(X') \cong \text{Ter}(X)$. It is unique because AtlTrav preserves orders and terminal because H_E is faithful.

10 Coalitions

Definition 10.1 (The Coalizing Functor). The **Coalizing Functor**, denoted

$$\text{Coa} : \text{StaAtlTrav} \longrightarrow \text{DomIns},$$

is the extent of the charted atlas: $\text{Coa} = \text{Ex} \circ \text{Chr}$.

Definition 10.2 (The Coalition of an Atlas). The **coalition** of an atlas X is the extent of its chart, regarding X as an object of the wide category of stable traversals.

Definition 10.3 (The Domanial Inclusion Functor). The **Domanial Inclusion Functor** $\text{DomInc} : \text{DomIns} \rightarrow \text{StaAtlTrav}$ sends a dominion to the atlas of cardinality 1 whose coalition is that dominion. It is left adjoint to Coa : for $X : \text{DomIns}$ and $Y : \text{StaAtlTrav}$, naturally in X and Y ,

$$\text{Hom}_{\text{StaAtlTrav}}(\text{DomInc}(X), Y) \cong \text{Hom}_{\text{DomIns}}(X, \text{Coa}(Y)),$$

and $\text{Coa}(\text{DomInc}(X)) \cong X$.

11 Data Transformations

Definition 11.1 (The Category of Data Transformations). The **Category of Data Transformations**, also called the category of **Data Transformation Sets** or **DaTra Sets**, is the presheaf category

$$\text{DaTra} = [\text{Atl}^{\text{op}}, \text{Set}].$$

As a presheaf category, it is a topos.

Definition 11.2 (Navigation). A **navigation** of $D : \text{DaTra}$ is a monomorphism $\text{Nav} : \text{Yo}(A) \rightarrow D$ for some atlas A .

Definition 11.3 (Expedition). An **expedition** is a navigation represented by an Atlas Map.

Definition 11.4 (The Category of Data Transposals). The **Category of Data Transposals**, denoted DaTrap , is the presheaf category $[\text{AtlTrap}^{\text{op}}, \text{Set}]$.

Definition 11.5 (The Category of Data Traversals). The **Category of Data Traversals**, denoted DaTrav , is the presheaf category $[\text{AtlTrav}^{\text{op}}, \text{Set}]$.

Definition 11.6 (Data Transformation Maps). The **Category of Data Transformation Maps**, denoted DaTraMap , is the full subcategory of DaTra Sets all of whose navigations are expeditions.

12 Atlas Federations

Definition 12.1 (Atlas Merge). The **Atlas Merge** is the object operation

$$\text{AtlMerge} : \text{Ob}(\text{Atl}) \times \text{Ob}(\text{Atl}) \longrightarrow \text{Ob}(\text{Atl}).$$

For atlases X and Y , the extent of $\text{AtlMerge}(X, Y)$ is the disjoint union of their extents. Both atlases are evaluated on their full spines: page $n + 1$ of the merge is the tagged, componentwise combination of page n of X and page n of Y .

Definition 12.2 (Atlas Federation). An **Atlas Federation** is a countable tagged family of atlas objects. The category of Atlas Federations is denoted

$$\text{AtlFed}.$$

A morphism consists of a map of tags and, at each source tag, a stable atlas traversal to its selected target tag.

Definition 12.3 (The Empty Atlas). The **Empty Atlas** is $\text{AtII} = \text{DomInc}(\emptyset)$, also viewed as an Atlas Federation.

Definition 12.4 (The Atlas Horizontal Sum Bifunctor). The **Atlas Horizontal Sum Bifunctor**, denoted $\text{AtlHorSum} : \text{AtlFed} \times \text{AtlFed} \rightarrow \text{AtlFed}$, has object action

$$\text{AtlHorSum}(X, X') = \text{AtlMerge}(X, X').$$

Here AtlFed is the category of Atlas Federations; its empty federation represents AtII . Given stable traversals $F : X \rightarrow Y$ and $F' : X' \rightarrow Y'$, horizontal sum preserves the left and right tags and applies F and F' componentwise. Stability fixes the common origin, so this arrow action is again a stable atlas traversal.

Lemma 12.5 (Horizontal Lemma). *The Atlas Horizontal Sum Bifunctor AtlHorSum exists.*

Proof. Identity arrows act identically on every tag, composition holds componentwise, and every component arrow is required to lie in StaAtI Trav . The federation structure supplies the associator, unitors, and braider by reassociation, deletion of the empty tag family, and tag swapping, respectively.

Definition 12.6 (The Atlas Braider). The **Atlas Braider** $\text{AtlBrd}_{F, F'} : \text{AtlHorSum}(F, F') \rightarrow \text{AtlHorSum}(F', F)$ swaps the left and right component tags. Hence $\text{AtlBrd}_{F, F'} \circ \text{AtlBrd}_{F', F} = \text{Id}$.

Definition 12.7 (The Atlas Associator). The **Atlas Associator**, denoted

$$\text{AtlAsoc}_{F, F', F''} : \text{AtlHorSum}(\text{AtlHorSum}(F, F'), F'') \longrightarrow \text{AtlHorSum}(F, \text{AtlHorSum}(F', F'')),$$

flattens the nested federation and retags its three sides using the canonical reassociation $((x + y) + z) \leftrightarrow (x + (y + z))$. Its data component is the matching reassociation of the three extent summands.

Definition 12.8 (The Atlas Unitors). The **Atlas Left Unitor** $\text{Lu} : \text{AtlHorSum}(\text{AtII}, F) \rightarrow F$ and the **Atlas Right Unitor** $\text{Ru} : \text{AtlHorSum}(F, \text{AtII}) \rightarrow F$ remove the empty tag family.

13 Stable Data Transversals

Definition 13.1 (Stable Data Transversals). The **Category of Stable Data Transversals**, denoted StaDaTrav , is the presheaf category on Atlas Federations whose component arrows lie in StaAtI Trav .

Definition 13.2 (The Empty Map). The **Empty Map**, denoted I , is the Day unit represented by the empty Atlas Federation:

$$I = \text{Yo}(\text{AtII}).$$

Definition 13.3 (The Horizontal Sum Bifunctor). The **Horizontal Sum Bifunctor**, denoted $\text{HorSum} : \text{StaDaTrav} \times \text{StaDaTrav} \rightarrow \text{StaDaTrav}$, or in infix notation by $\boxplus : \text{StaDaTrav} \times \text{StaDaTrav} \rightarrow \text{StaDaTrav}$, is the Day convolution extension of stable atlas horizontal sum. Because the indexing morphisms are arrows of StaAtI Trav , its result and its arrow action land in StaDaTrav by construction.

Definition 13.4 (The Stable Data Transversal Braider). The **Stable Data Transversal Braider** $\text{Brd}_{F, F'} : F \boxplus F' \rightarrow F' \boxplus F$ is the Day convolution extension of AtlBrd .

Definition 13.5 (The Stable Data Transversal Associator). The **Stable Data Transversal Associator**

$$\text{Asoc}_{F, F', F''} : (F \boxplus F') \boxplus F'' \longrightarrow F \boxplus (F' \boxplus F'')$$

is the Day convolution extension of AtlAsoc .

Definition 13.6 (The Stable Data Transversal Unitors). The **Stable Data Transversal Left Unitor** $\text{Lu} : I \boxplus F \rightarrow F$ and the **Stable Data Transversal Right Unitor** $\text{Ru} : F \boxplus I \rightarrow F$ are the Day convolution extensions of AtlLu and AtlRu .

Definition 13.7 (Stable Data Transversals Monoidal Category). The **Stable Data Transversals Monoidal Category**, denoted StaDaTravMon , is

$$\text{StaDaTravMon} = (\text{StaDaTrav}, \boxplus, I).$$

Its tensor is Day convolution on the Atlas Federation. The braider, associator, and unitors are induced by retagging that form, so all coherence maps remain within stable data transversals.